Behind the Paycheck:

What DC Government Salaries Reveal About Public Service

MATH 014 — Introduction to Data Science | Spring 2026
Howard University
Dataset: DC Public Employee Salary — DC Open Data
Submitted: April 29, 2026

01 | Objective

The objective of this project is to explore and analyze the DC Public Employee Salary dataset to uncover patterns in how the District of Columbia compensates its workforce. Specifically, this project aims to understand the distribution of annual salaries across the DC government, identify which agencies employ the most workers and pay the highest wages, examine how appointment types vary across the workforce, and analyze how hiring activity has changed over time. By answering these questions, this analysis seeks to provide a clearer picture of the structure, scale, and equity of public sector employment in Washington, DC.

02 | Introduction

The DC Public Employee Salary dataset is published by the District of Columbia government through the DC Open Data portal. It contains compensation and employment records for thousands of current DC government employees across a wide range of agencies, job titles, and appointment types. Each record includes an employee's first and last name, job title, the agency they work for, their pay grade, their compensation rate (either annual salary or hourly rate), their hire date, and their type of government appointment.

This dataset is valuable for several reasons. First, as a publicly funded institution, the DC government has a transparency obligation to its residents, and this data fulfills that commitment by making compensation information publicly accessible. Second, salary data offers a window into the priorities of a government — which sectors it staffs most heavily, which roles it values most financially, and how its workforce has grown or

shifted over time. Third, for students of data science, this dataset presents a realistic and rich challenge: it contains over 40,000 employee records, mixed data types, and values that require careful interpretation, such as distinguishing annual salaries from hourly rates.

After removing hourly-rate records and focusing exclusively on annual salaries, the working dataset contains 35,627 employee records spanning dozens of agencies and hundreds of unique job titles. This report presents the findings of an exploratory data analysis of that dataset, with a focus on salary distributions, agency comparisons, appointment types, and hiring trends.

03 | Method

Data Understanding

The dataset contains 40,611 total records and 9 columns: FIRST_NAME, LAST_NAME, JOBTITLE, DESCRSHORT (agency abbreviation), GRADE (pay grade), COMPRATE (compensation rate), HIREDATE_STRING (hire date as a string), GVT_TYPE_OF_APPT (type of government appointment), and OBJECTID (a unique record identifier).

Data Cleaning

Several cleaning steps were necessary before analysis could begin:

- Type Conversions: The COMPRATE column was stored as a string and was converted to a numeric float. The HIREDATE_STRING column was converted to a datetime object using pandas, which allowed for year extraction and trend analysis. The GRADE column was also converted from string to numeric.
- Handling Mixed Compensation Rates: A key discovery during data understanding was that the COMPRATE column contained two distinct types of values — annual salaries (ranging from $10,000 to $390,232) and hourly wage rates (small decimal values, such as $41.73 per hour). These two types of records cannot be analyzed together in a meaningful salary analysis. A threshold of $500 was applied: records with a COMPRATE below $500 were classified as hourly-rate employees (4,971 records) and excluded from the main salary analysis. The remaining 35,627 records, which represent annually salaried employees, formed the primary dataset.
- Missing Values: The GRADE column contained 40 missing (null) values after numeric conversion. These records were retained for all analyses that did not require grade information, and excluded from the grade-specific analysis.
- Duplicate Check: No exact duplicate records were identified in the dataset. Each OBJECTID value was unique, confirming record integrity.
- Column Derived: A HIRE_YEAR column was extracted from HIREDATE_STRING to support the hiring trend analysis over time.

After cleaning, the analysis focused on 35,627 annual-salary records, which represent the core salaried workforce of the DC government.

04 | Storytelling: Data Visualization and Interpretation

Figure 1: Distribution of Annual Salaries

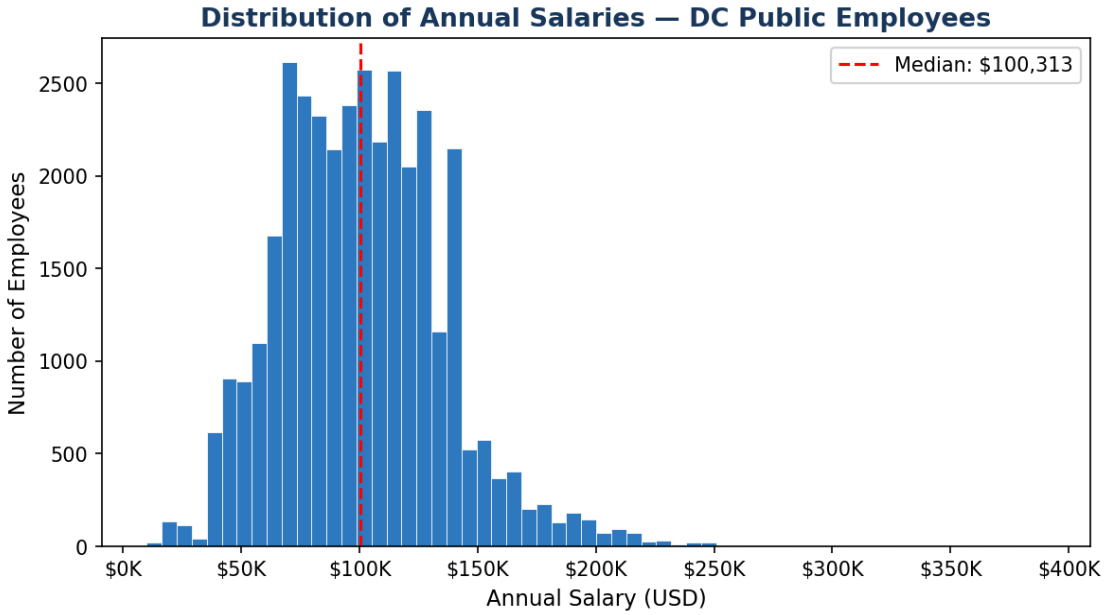

Figure 1. Histogram showing the distribution of annual salaries across 35,627 DC government employees. The red dashed line marks the median salary of $100,313.

Figure 1 presents the overall distribution of annual salaries for DC government employees. The histogram reveals a right-skewed distribution, meaning that the majority of employees earn salaries clustered in the lower-to-mid range, while a smaller number of high-earning employees pull the mean upward. The median salary is $100,313, while the mean is $101,544 — a small difference that confirms the distribution is approximately symmetric at its center but has a long right tail reaching up to $390,232.

The most common salary range falls between roughly $75,000 and $130,000, reflecting a large middle tier of career service employees. The presence of salaries as low as $10,000 likely reflects part-time or entry-level positions, while the highest salaries correspond to executive-level roles and senior management positions. This spread illustrates meaningful income variation within a single government workforce and raises questions about what factors — such as agency, grade, or appointment type — drive salary differences.

Figure 2: Average Salary by Agency

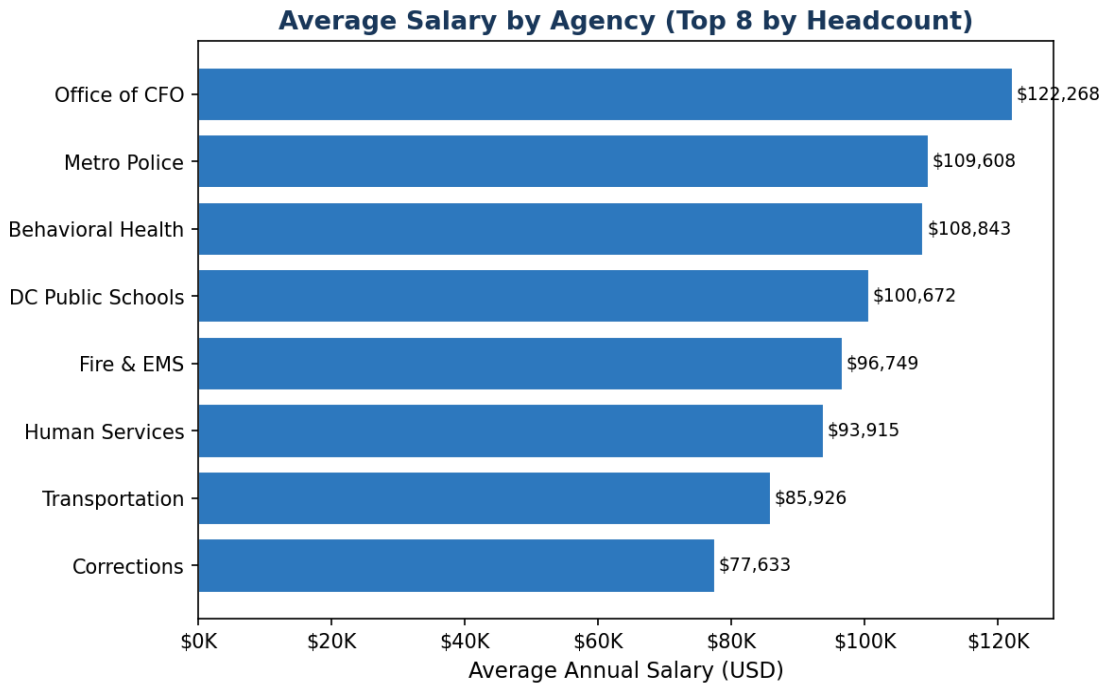

Figure 2. Horizontal bar chart showing average annual salary for the eight largest agencies by employee headcount.

Figure 2 compares average annual salaries across the eight agencies with the largest headcounts. The Office of the Chief Financial Officer (OCFO) leads with an average salary of approximately $122,268, followed by the Metropolitan Police Department (MPD) at $109,608 and the Department of Behavioral Health (DBH) at $108,843. At the other end, the Department of Corrections (DOC) averages $77,633 — a difference of over $44,000 compared to OCFO employees.

These differences reflect the professional composition of each agency. OCFO employs large numbers of financial analysts, accountants, and auditors — roles that command higher market wages both in the public and private sectors. MPD's elevated average reflects police officer salaries, which include premium pay and overtime compensation. DC Public Schools (DCPS), the largest employer in the dataset with over 10,000 employees, shows an average salary of $100,672 — a figure that incorporates both classroom teachers and administrative staff. The Department of Transportation (DDOT) ranks second to last among these agencies at $85,926, likely because it employs a larger share of skilled trades and technical workers at lower grade levels.

Figure 3: Appointment Type Distribution

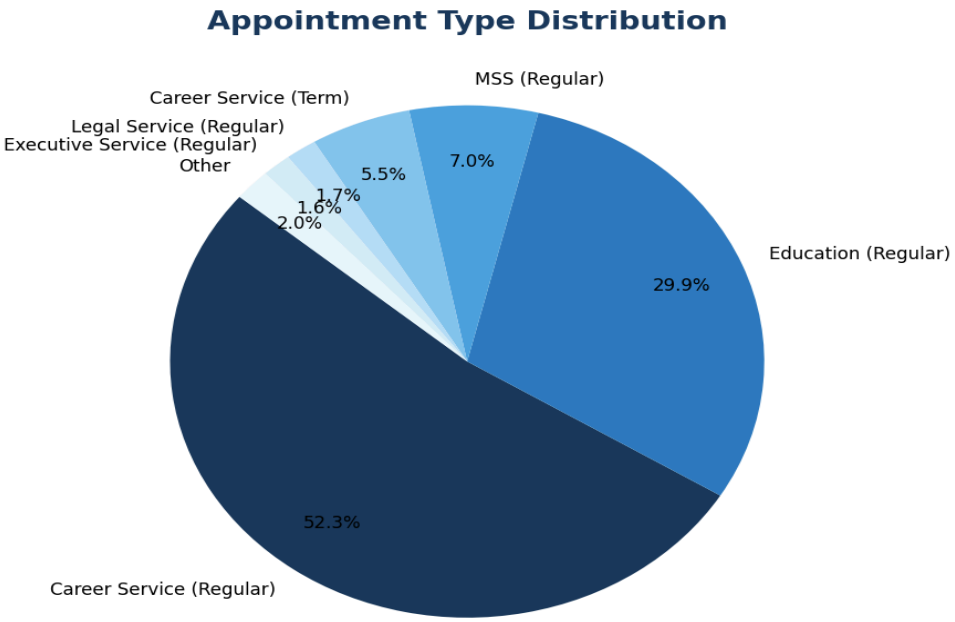

Figure 3. Pie chart showing the breakdown of government appointment types among annually salaried employees.

Figure 3 displays how employees are distributed across different types of government appointments. Career Service Regular (CS - Reg) is the most common appointment type, accounting for 52.3% of all salaried employees — this represents the core civil service workforce with full employment protections. Education Regular (Ed - Reg) makes up 29.9% of the workforce, reflecting the dominance of DCPS with its 10,133 employees. Management Supervisory Service Regular (MSS - Reg), Career Service Term (CS - Term), and Legal Service Regular (LS - Reg) together account for most of the remaining share.

A notable finding is that term and temporary appointment types — CS - Term, CS - Temp, Ed - Temp, MSS - Temp — collectively account for approximately 6.5% of the salaried workforce. These employees work under time-limited contracts without the same job security as regular-status employees. This raises important questions about workforce stability and whether DC government agencies rely on short-term staff for ongoing functions that would typically be filled by permanent employees.

Figure 4: Hiring Trend Over Time

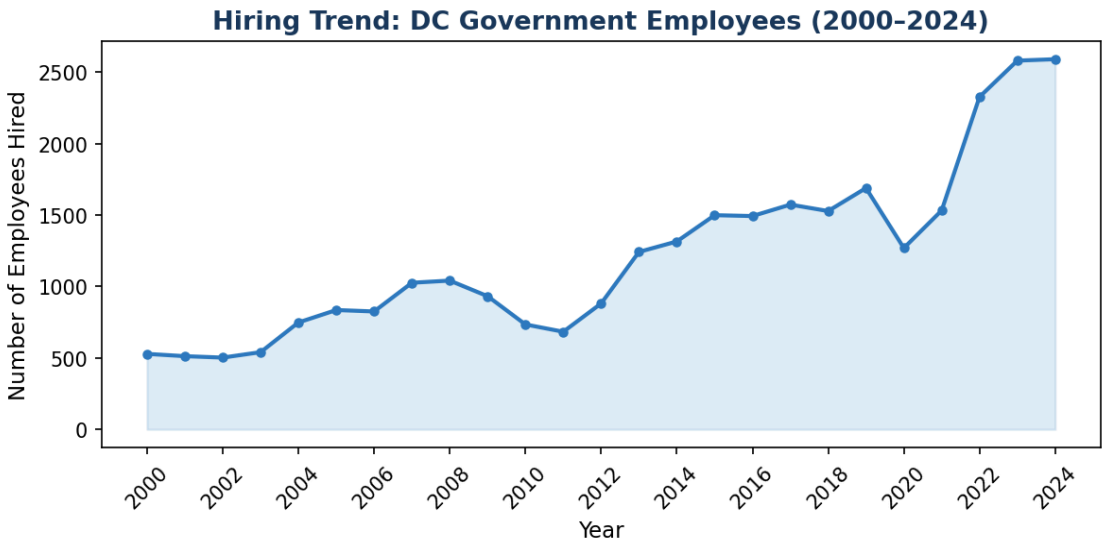

Figure 4. Line chart showing the number of employees hired per year by the DC government from 2000 to 2024.

Figure 4 tracks the number of DC government employees hired each year between 2000 and 2024. The trend shows significant variation, with hiring peaking in the mid-2000s and again more recently. A notable dip in hiring occurs around 2010 to 2013, likely reflecting the fiscal constraints that many state and local governments experienced in the aftermath of the 2008 financial crisis. Hiring accelerated in subsequent years, with the largest spikes occurring around 2020 and 2021.

The surge in hiring around 2020 is particularly striking and may reflect DC's response to the COVID-19 pandemic, which required rapid expansion of public health, emergency services, and social support infrastructure. The continued strong hiring in 2021 through 2023 suggests sustained workforce growth in the recovery period. Overall, the trend confirms that the DC government has been actively expanding its workforce over the past two decades, with the current dataset primarily reflecting employees hired in the 2010s and 2020s.

05 | Conclusion

This analysis of the DC Public Employee Salary dataset reveals several important truths about the structure of Washington, DC's public workforce. The median annual salary of $100,313 places most DC government workers solidly in a middle-income range, though the distribution's long right tail — with some employees earning well over $300,000 — indicates that executive-level compensation is substantially higher than the typical employee's pay. The agency-level comparison shows that financial and law enforcement roles command higher average salaries, while positions in corrections and transportation tend to fall at the lower end of the pay spectrum.

The appointment type analysis reveals that over half of DC's salaried workforce holds career service regular status, suggesting a relatively stable core workforce. However,

the meaningful presence of term and temporary employees raises questions about job security and long-term workforce planning. The hiring trend analysis captures a clear narrative: the DC government contracted its hiring after the 2008 financial crisis, then expanded significantly through the 2010s and 2020s, with a notable spike that may be tied to pandemic-era staffing needs.

Taken together, these findings suggest that DC government employment is relatively well-compensated but varied — shaped by the missions of individual agencies, the grade levels of positions filled, and the employment conditions under which workers are hired. Future analysis could explore salary equity across demographic groups (if such data were available), the relationship between tenure and salary growth, or a more granular comparison of job titles across agencies. The DC Public Employee Salary dataset, despite its relatively straightforward structure, holds rich stories about how a major American city invests in its public servants.

06 | References

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